a) (2.5×2) hours

b) $\left(\frac{2.5}{2}\right)$ hours

c) 2.5 hours

d) Without knowing the rate constant, $t_{1/2}$ cannot be determined from the given data

5. For the reaction, $2NH_3 \longrightarrow N_2 + 3H_2$, if $\frac{-d[NH_3]}{dt} = k_1[NH_3]$, $\frac{d[N_2]}{dt} = k_2[NH_3]$, $\frac{d[H_2]}{dt} = k_3[NH_3]$ then the relation between k_1, k_2 and k_3 is

a) $k_1 = k_2 = k_3$

b) $k_1 = 3\,k_2 = 2\,k_3$

c) $1.5\,k_1 = 3\,k_2 = k_3$

d) $2k_1 = k_2 = 3\,k_3$

6. The decomposition of phosphine (PH_3) on tungsten at low pressure is a first order reaction. It is because the (NEET)

a) rate is proportional to the surface coverage

b) rate is inversely proportional to the surface coverage

c) rate is independent of the surface coverage

d) rate of decomposition is slow

7. For a reaction Rate $= k[\text{acetone}]^{3/2}$ then unit of rate constant and rate of reaction respectively is

a) $(\text{mol L}^{-1}\text{s}^{-1}), (\text{mol}^{-1/2}\ \text{L}^{1/2}\ \text{s}^{-1})$

b) $(\text{mol}^{-1/2}\ \text{L}^{1/2}\ \text{s}^{-1}), (\text{mol L}^{-1}\text{s}^{-1})$

c) $(\text{mol}^{1/2}\ \text{L}^{1/2}\ \text{s}^{-1}), (\text{mol L}^{-1}\text{s}^{-1})$

d) $(\text{mol L s}^{-1}), (\text{mol}^{1/2}\ \text{L}^{1/2}\text{s})$

8. The addition of a catalyst during a chemical reaction alters which of the following quantities? (NEET)

a) Enthalpy b) Activation energy

c) Entropy d) Internal energy

9. Consider the following statements :

(i) increase in concentration of the reactant increases the rate of a zero order reaction.

(ii) rate constant k is equal to collision frequency A if $E_a = 0$

(iii) rate constant k is equal to collision frequency A if $E_a = \infty$

(iv) a plot of $\ln(k)$ vs T is a straight line.

(v) a plot of $\ln(k)$ vs $\left(\frac{1}{T}\right)$ is a straight line with a positive slope.

Correct statements are

a) (ii) only b) (ii) and (iv)

c) (ii) and (v) d) (i), (ii) and (v)

10. In a reversible reaction, the enthalpy change and the activation energy in the forward direction are respectively $-x$ kJ mol^{-1} and y kJ mol^{-1}. Therefore, the energy of activation in the backward direction is

a) $(y - x)$ kJ mol^{-1}

b) $(x + y)$ J mol^{-1}

c) $(x - y)$ kJ mol^{-1}

d) $(x + y) \times 10^3$ J mol^{-1}

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